

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 322 — Past Questions & Materials
by Renance

Course: MTH 322
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

This study pack was compiled and restructured by Renance for MTH 322 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 2 — MTH 322, 311. 2019.

Level 300 • First Semester • PastQuestion

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is the content of the following polynomials $3x^2 + 6x + 12$ and 2°

Let p be a prime number, Show $x^4 + 1$ is irreducible

that $f(x)$

2. Let ϕ be a Group homomorphism (i) Group Isomorphism (i) Cycle subgroup is a

homomorphism, prove the following (i) $f(e_i) = e_i$ where e_i is the identity of G_i for

(ii) $f(x) = [f(x)]$ for all x in G . Show that f is a homomorphism

Let $f: (C, +) \rightarrow (R, +)$ by $f(x) = 2x$, then f is a homomorphism. What are Image of f and Kernel of f ?

(i) Even permutation (ii) Disjoint cycles (iii) Permutation of n elements is symmetric group

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(5 bits 35% =) (i) Express it as a product of disjoint cycles (ii) Find its -

Express it as a product of trans

2 16.78

Positions (iv) Verify if it commutes with the permutation

without proof

Let G be a group of order 30 either has a normal sub

$H = \{0, 5\}$ and $K = \{0, 2, 4, 6, 8\}$

group G be the internal direct product of its subgroups H and K then prove that each $x \in G$ can be expressed as $x = hk$ where $h \in H, k \in K$ and that $hk = kh \forall h \in H, k \in K$.

group of order 5 or a normal subgroup of order 3

(i) Euclidean domain (ii) Principal Ideal domain (iii) Unique factorization domain

& (the set of all integers) is a Euclidean domain. (ii) Find all the units in $R = \{a + b\sqrt{5} : a, b \in \mathbb{Z}\}$

the greatest common divisor of two integers (g.c.d.) and find the g.c.d. of the following (ii) $p(x) =$

every subgroup of a commutative group is normal

that the subgroup $\langle (2 \ 3) \rangle$ of the Symmetric group S_3 is not normal

Let G be a group of order 10

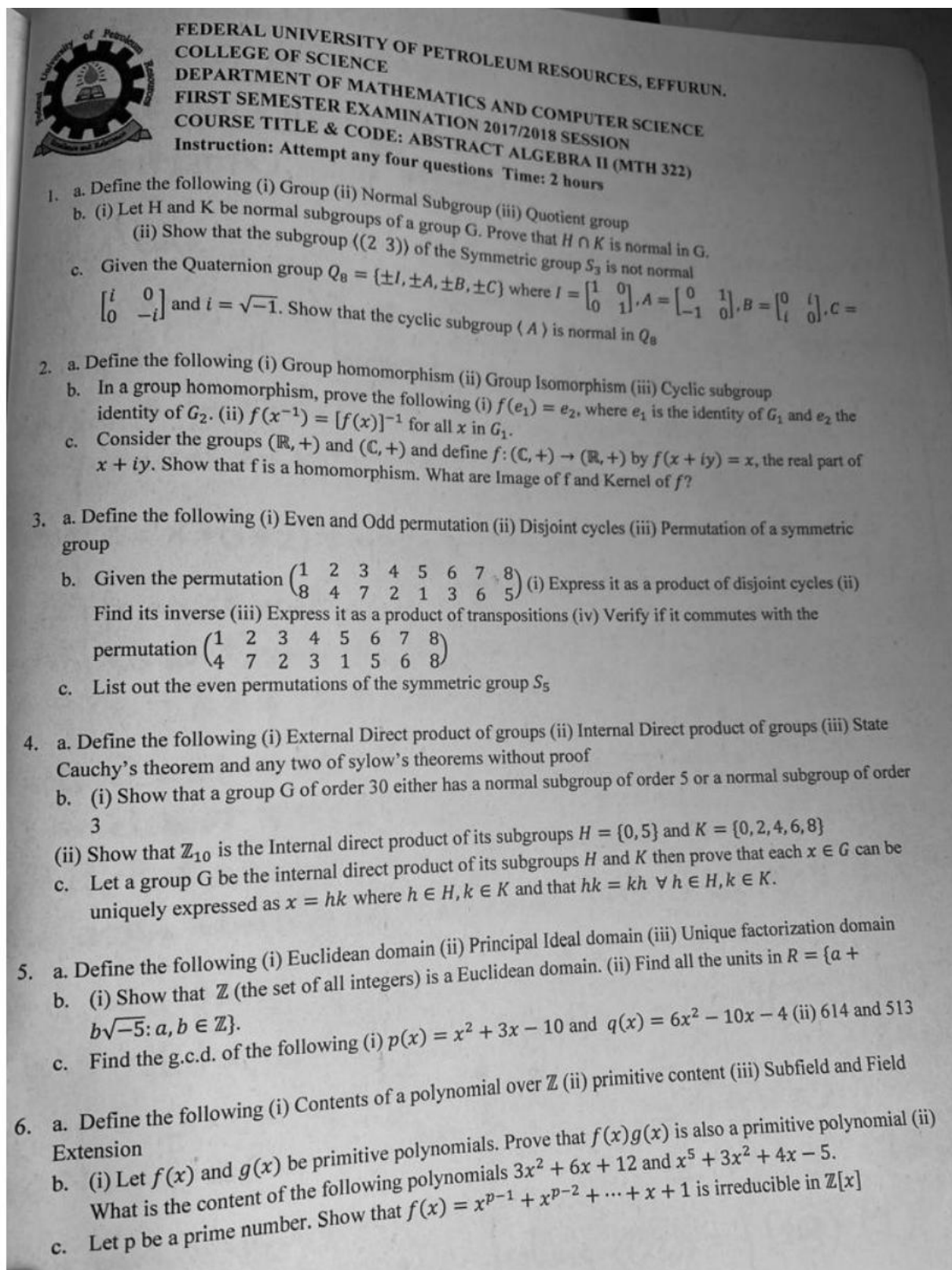
Let G be a group of order 10. Let $H = \{e, a, a^2, a^3, a^4\}$ be a subgroup of G . Show that H is normal in G .

Let G be a group of order 10. Let $H = \{e, a, a^2, a^3, a^4\}$ be a subgroup of G . Show that H is normal in G .

Paper 2 of 2 — MTH 322, 2018

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2017/2018 SESSION
COURSE TITLE & CODE: ABSTRACT ALGEBRA II (MTH 322)
Instruction: Attempt any four questions Time: 2 hours

- Define the following (i) Group (ii) Normal Subgroup (iii) Quotient group
 - (i) Let H and K be normal subgroups of a group G . Prove that $H \cap K$ is normal in G .
(ii) Show that the subgroup $\langle (2 \ 3) \rangle$ of the Symmetric group S_3 is not normal
 - Given the Quaternion group $Q_8 = \{\pm I, \pm A, \pm B, \pm C\}$ where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$, $C = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$ and $i = \sqrt{-1}$. Show that the cyclic subgroup $\langle A \rangle$ is normal in Q_8
- Define the following (i) Group homomorphism (ii) Group Isomorphism (iii) Cyclic subgroup
 - In a group homomorphism, prove the following (i) $f(e_1) = e_2$, where e_1 is the identity of G_1 and e_2 the identity of G_2 . (ii) $f(x^{-1}) = [f(x)]^{-1}$ for all x in G_1 .
 - Consider the groups $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$ and define $f: (\mathbb{C}, +) \rightarrow (\mathbb{R}, +)$ by $f(x + iy) = x$, the real part of $x + iy$. Show that f is a homomorphism. What are Image of f and Kernel of f ?
- Define the following (i) Even and Odd permutation (ii) Disjoint cycles (iii) Permutation of a symmetric group
 - Given the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 4 & 7 & 2 & 1 & 3 & 6 & 5 \end{pmatrix}$ (i) Express it as a product of disjoint cycles (ii) Find its inverse (iii) Express it as a product of transpositions (iv) Verify if it commutes with the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 7 & 2 & 3 & 1 & 5 & 6 & 8 \end{pmatrix}$
 - List out the even permutations of the symmetric group S_5
- Define the following (i) External Direct product of groups (ii) Internal Direct product of groups (iii) State Cauchy's theorem and any two of sylow's theorems without proof
 - (i) Show that a group G of order 30 either has a normal subgroup of order 5 or a normal subgroup of order 3
(ii) Show that $\mathbb{Z}_{10}$ is the Internal direct product of its subgroups $H = \{0, 5\}$ and $K = \{0, 2, 4, 6, 8\}$
 - Let a group G be the internal direct product of its subgroups H and K then prove that each $x \in G$ can be uniquely expressed as $x = hk$ where $h \in H, k \in K$ and that $hk = kh \ \forall h \in H, k \in K$.
- Define the following (i) Euclidean domain (ii) Principal Ideal domain (iii) Unique factorization domain
 - (i) Show that $\mathbb{Z}$ (the set of all integers) is a Euclidean domain. (ii) Find all the units in $R = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$.
 - Find the g.c.d. of the following (i) $p(x) = x^2 + 3x - 10$ and $q(x) = 6x^2 - 10x - 4$ (ii) 614 and 513
- Define the following (i) Contents of a polynomial over $\mathbb{Z}$ (ii) primitive content (iii) Subfield and Field Extension
 - (i) Let $f(x)$ and $g(x)$ be primitive polynomials. Prove that $f(x)g(x)$ is also a primitive polynomial (ii) What is the content of the following polynomials $3x^2 + 6x + 12$ and $x^5 + 3x^2 + 4x - 5$.
 - Let p be a prime number. Show that $f(x) = x^{p-1} + x^{p-2} + \dots + x + 1$ is irreducible in $\mathbb{Z}[x]$